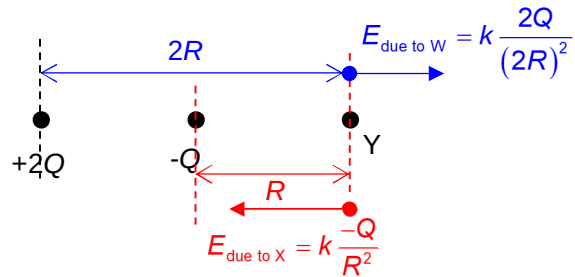


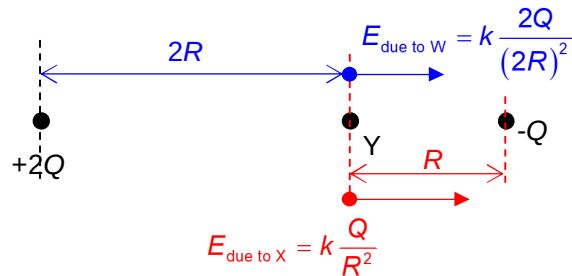
20 take right direction as +ve:

Initial:



$$E_{y, \text{initial}} = k \frac{2Q}{(2R)^2} - k \frac{Q}{R^2} = -k \left(\frac{Q}{2R^2} \right)$$

final:



$$E_{y, \text{final}} = k \frac{2Q}{(2R)^2} + k \frac{Q}{R^2} = k \left(\frac{3Q}{2R^2} \right)$$

E increased by factor of 3 times.

potential is scalar; since Y is equidistant from X and Z , no change in total potential.

21 by potential divider rule